A User Experience Study of Pain Points in the KMITL Registration System for Interface Redesign

Jeerawiwat Wihatanone
Student ID 66010133

Pattarapong Hongkanjanagul
Student ID 66010615

Nattichaporn Peakkuntod
Student ID 66011357

Department of IoT and Information Engineering, Faculty of Engineering
King Mongkut's Institute of Technology Ladkrabang (KMITL), Bangkok, Thailand
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***Abstract*—Course registration is a time-critical task that every student performs each semester. From their own experience the authors suspected that the KMITL registration system (REG) makes it hard to find suitable sections, shows seat and exam information unclearly and reveals schedule conflicts late; these suspicions were not verified by a survey. We designed a prototype of a new registration page (search and filters, separate required and general-education tabs, immediate conflict alerts that name the clashing day, time or exam date, a review step, and a history of additions and withdrawals) and evaluated it in moderated think-aloud sessions with 15 participants (13 KMITL students, 2 non-students) who performed seven tasks. Mean task success was 87.6%, but two tasks fell below the 80% criterion: finding a GenEd language course (T3, 60%) and adding a course with an exam-date clash (T5, 66.7%). Students needed a mean of 4:37.1 min (SD 59.7 s) for all seven tasks together. Mean Single Ease Question (SEQ) ratings of the 13 students ranged from 4.08 to 4.85 out of 5 per task, and the mean System Usability Scale (SUS) score was 75.6 (SD 17.1; 95% CI 65.3–85.9), above the benchmark of 68. Eleven usability problems were identified from the comments of eight participants and from observation, most often the lack of GenEd course-type labels and clear categories, an unclear "available seats" filter and small class-time text. The prototype was not compared with the current system, and success counts pool students and non-students, so the results describe this sample only.**

***Keywords*—user experience, usability testing, course registration, System Usability Scale, interface redesign**

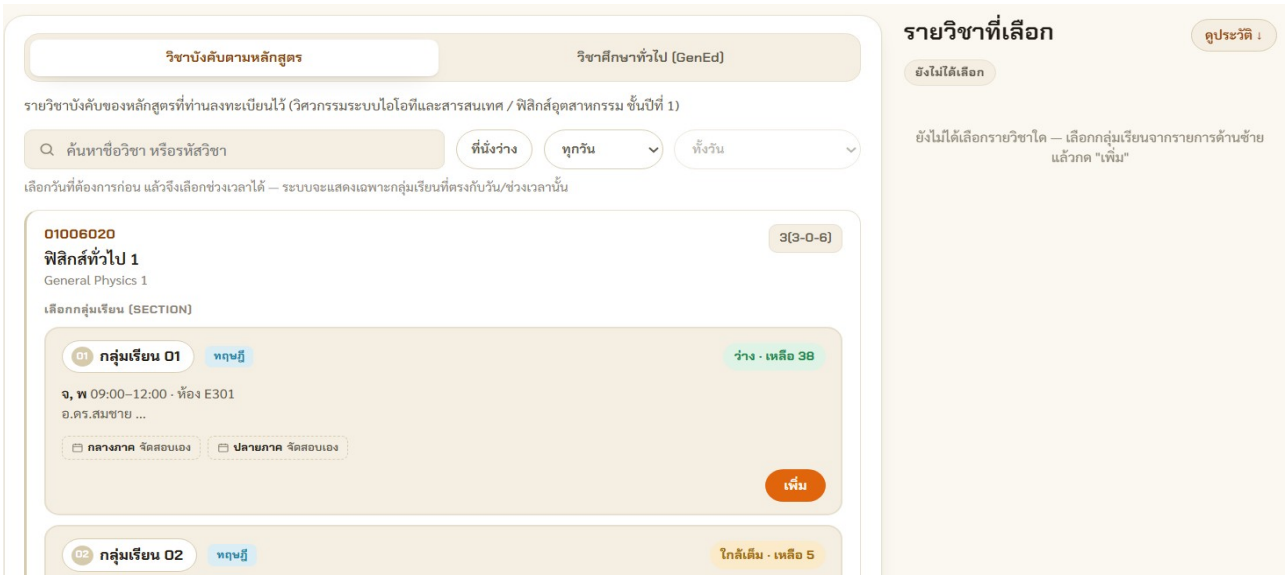


Fig. 1. The redesigned registration prototype (mock data). Left: required/GenEd tabs, search box, day and period filters and a section card showing seats left; right: the "Selected courses" box. The page is labelled as a student mock-up and is not the official REG site.

I. INTRODUCTION

Every KMITL student uses the REG registration system to register for courses, check results and follow the timetable. Registration is the most time-critical of these activities: within a limited window students must choose a section for each course and must avoid clashes of class day, class time or exam date. Mistakes often have to be repaired later by adding or withdrawing courses, which can disturb the study plan.

From their own experience with REG, the authors suspected four pain points: (i) finding a section that fits one's timetable is hard because the list cannot be filtered by day, time or available seats; (ii) timetable, seat and exam information is displayed unclearly; (iii) clashes are noticed late; and (iv) error messages are hard to understand. These are the authors' observations. They were not confirmed by a survey or by testing REG itself, so in this paper they serve as design hypotheses, not findings.

To explore these hypotheses we designed a prototype of a new registration page (Fig. 1) and evaluated it with 15 participants in moderated think-aloud sessions. We ask three research questions:

RQ1 What share of seven core registration tasks can users complete on the prototype, and how long do they take?

RQ2 How easy and satisfying do users find the prototype (SEQ and SUS)?

RQ3 Which tasks cause the most trouble, and which screen elements are responsible?

The contributions are (1) a prototype whose six features are tied to Nielsen's heuristics, (2) usability evidence from 15 participants: task success and errors, total task time, per-task ease ratings, SUS scores and eleven usability problems, and (3) a candid account of what the data can and cannot support. In particular, RQ1 is answered with success rates

pooled over all 15 participants and with total time only, because per-task times were not recorded.

II. RELATED WORK

A. Usability, heuristics and measurement

ISO 9241-11 defines usability as the effectiveness, efficiency and satisfaction with which specified users achieve specified goals in a specified context [1]. We map task success and errors to effectiveness, time to efficiency, and SEQ and SUS to satisfaction. Nielsen's ten heuristics [2] guided the design and the classification of problems (H1–H10 below). SUS [3] is a ten-item questionnaire scored 0–100; across many studies its mean is about 68 [4], which we use as a reference point. SEQ asks one question about the ease of a task and is normally rated on seven points [5]; we adapted it to five points and collected it after all tasks, so our scores cannot be compared with published SEQ benchmarks. Think-aloud protocols [6] expose why users hesitate. Testing about five users finds most problems [7], [8], but that figure concerns problem discovery, not the precision of quantitative estimates.

B. Course registration systems

Estevez et al. [9] studied the online registration system of a Canadian university: they examined the existing system, built a prototype, tested it and compared it with the original. The prototype improved usability and satisfaction, and the study showed that some students believed they had registered when they had only added courses to a cart, which makes clear status feedback important. Ngamassi et al. [10] surveyed students about a university registration system; 50 of 150 invited students responded and overall satisfaction was 2.4 out of 5.

C. Studies of icons and visual elements

Work from our own institution examines how the visual properties of interface elements affect selection. Satcharoen [11] tested six icon sizes (16–96 px) and found that larger icons tended to be identified faster and more accurately, although some differences were not significant. A follow-up on mobile devices (n = 52) found the highest inaccuracy for small icons (48 px and smaller) and higher inaccuracy at longer selection times [12]. Among 400 undergraduates, high icon entropy lowered accuracy, but significantly so only under the strictest time limit of 5 s [13]. Satcharoen et al. [14] redesigned icons to reduce confusion and collected user judgments through web questionnaires. Together these studies suggest that element size, visual complexity, clarity of meaning and time pressure all shape selection performance, and we use them as a lens in Section V.

III. METHOD

A. Prototype

The prototype is an HTML page that simulates registration with mock data (30 required and 56 general-education courses) and is not connected to the real system; it is available at pattaraponghongkanjanagul.github.io/reg-redesign. Six features address the hypothesised pain points: (1) search by course name or code with filters for day, time period and available seats (H7); (2) separate tabs for required and GenEd courses, with the same section card in both (H4), showing day, time, room, instructor, seats left and exam date (H6); (3) a "Selected courses" box with total credits and an always-visible confirm button (H1); (4) an immediate alert when a chosen section clashes, naming the clashing class day, class time or exam date and the other course (H5, H9); (5) a review step before confirmation, with editing possible afterwards (H3, H5); and (6) a time-stamped history of additions and withdrawals (H1, H3). Three author-defined personas (a final-year student who wants a fast, clash-free timetable; a final-year student who worries about missing a step and not knowing whether registration succeeded; and a graduate unfamiliar with terms such as GenEd and HyFlex) guided the features and tasks.

B. Participants

Participants were recruited by convenience sampling. Students were required to have used REG for at least one semester. Fifteen people took part: 13 KMITL students and 2 non-students (P01 and P10) who did not meet this criterion and are reported separately. All 15 returned the questionnaire (P01–P15). Of the 13 students, 12 used REG every semester and P02 occasionally; P01 used it occasionally and P10 never.

C. Tasks

Every participant performed the seven tasks in Table I in the same order, because T6 and T7 depend on the courses chosen earlier. Task wording did not reveal where buttons or menus were.

TABLE I
TEST TASKS AND THE FEATURE EACH EXERCISES

Task	Scenario	Feature tested
T1	Find Calculus 2 on Monday morning and add it.	Search, day/period filter
T2	Choose a General Physics 1 section that is not on Friday and still has seats.	Day and seat filters
T3	Find one GenEd language course held on Saturday that can be taken online.	GenEd tab, filters
T4	Add Introduction to Computer Programming (section 01), then another course held on Monday afternoon.	Class-time clash alert
T5	Add Academic English, then Japanese (section 01).	Exam-date clash alert
T6	Confirm all selected courses.	Confirm, review step
T7	Withdraw one course after confirming and check that the history shows it.	Editing, history

D. Procedure and measures

Sessions were individual and ran over a voice or video call on several platforms, each lasting about 3–12 minutes. Participants used a computer, an iPad or a mobile phone. The moderator explained that the prototype, not the participant, was being tested, obtained consent, explained think-aloud and let the participant explore the page freely. Tasks were then read aloud one by one. Timing started after T1 had been read and stopped when the participant said T7 was done, so only the total time for all seven tasks was recorded; per-task times were not. The moderator judged each task as completed independently, completed with help, or not completed, and noted errors such as failing to find a control. Sessions were recorded with consent. Afterwards participants completed a Google Forms questionnaire with: background (student status, frequency of REG use); SEQ on a 5-point scale (1 = very difficult, 5 = very easy) for each task; a 1–5 clarity rating of the class-time (T4) and exam-date (T5) clash messages; the 10 SUS items; a multiple-choice list of favourite features; and open-ended comments.

E. Analysis

The study is formative and the analysis descriptive. For time we report mean, SD, median and range per group, with students as the main group. SEQ is reported as mean (SD) per task. SUS was scored per respondent by the standard formula [3]; we report the student mean, SD and a 95% t-based confidence interval, and a sensitivity analysis that excludes respondents who gave the same answer to every item. Open-ended comments were grouped into usability problems, each linked to a screen element, the number of participants reporting it and a heuristic. A task was to count as troublesome if its success rate was below 80%, its mean SEQ below 4, or at least half of the participants made an error in it. Success rates count independent completions only and pool all 15 participants, because the records do not separate the groups. Participants were identified only by codes (P01–P15), took part voluntarily with informed consent, could withdraw at any time, and used mock data, so no real registration records were accessed.

IV. RESULTS

A. Task success and time (RQ1)

Table II gives task success. T1, T2, T4 and T7 were completed independently by all 15 participants. T3 (GenEd language course on Saturday, online) had the lowest success, 60% (9 of 15; five needed help and one failed), followed by T5 (66.7%; four needed help, one failed) and T6 (86.7%; two needed help). Mean success over the seven tasks was 87.6%, so T3 and T5 fall below the 80% criterion. Errors occurred only in three tasks: six in T3 (GenEd course not found; the term HyFlex unknown), five in T5 (academic English not found or not told apart from general English) and two in T6 (confirm button not found), involving 40%, 33.3% and 13.3% of participants. The other four tasks (T1, T2, T4, T7) had no errors.

TABLE II
TASK SUCCESS AND ERRORS (n = 15)

Task	Success (%)	Help / failed (n)	Erred ≥ 1 time (%)	Errors (n)
T1	100	0 / 0	0	0
T2	100	0 / 0	0	0
T3	60.0	5 / 1	40.0	6
T4	100	0 / 0	0	0
T5	66.7	4 / 1	33.3	5
T6	86.7	2 / 0	13.3	2
T7	100	0 / 0	0	0

Errors are counted over all 15 participants. Success counts independent completions only; students and non-students are pooled.

Table III and Fig. 2 show the total time for all seven tasks. Students took a mean of 4:37.1 min (SD 59.7 s; median 4:24.0; range 3:25.7–6:22.3), which would be roughly 40 s per task if the time were spread evenly (an indication only, since tasks were not timed separately). Two students needed more than six minutes. The two non-students took 6:34.0 and 12:25; including them raises the mean of all 15 to 5:16.1 and the SD to 134.3 s, while the median moves only from 4:24.0 to 4:35.2. Because only totals were timed and participants thought aloud, we cannot say which tasks were slow.

TABLE III
TOTAL TIME FOR THE SEVEN TASKS (MIN:S)

Statistic	Students (n = 13)	Non-students (n = 2)	All (n = 15)
Mean	4:37.1	9:29.5	5:16.1
Median	4:24.0	–	4:35.2
SD (s)	59.7	–	134.3
Minimum	3:25.7	6:34.0	3:25.7
Maximum	6:22.3	12:25.0	12:25.0

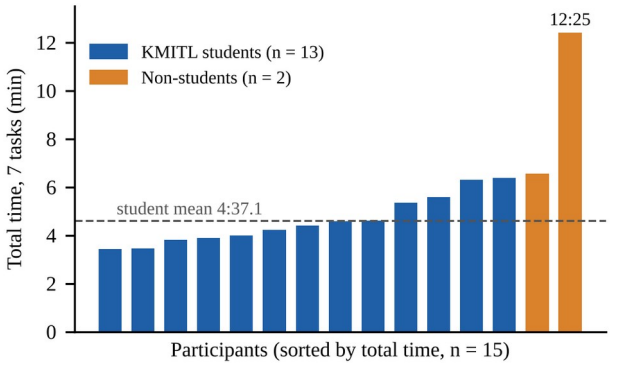


Fig. 2. Total time for all seven tasks, per participant (n = 15). The dashed line is the student mean.

B. Ease ratings (RQ2)

Table IV and Fig. 3 give the SEQ ratings. Students rated every task at 4.08 or higher, so no task fell below the threshold of 4. T7 (withdraw and check history) was rated easiest (4.85, SD 0.38); T3 was lowest and most variable (4.08, SD 1.19), followed by T4 (4.23, SD 1.24) and T2 (4.38). The wide SDs of T3–T5 (about 1.2) partly reflect P13, who gave T2 and T3 a 2 and T4 and T5 a 1. The ratings are close to the top of a 5-point scale and were given after all tasks, so they indicate relative difficulty at most; T5 shows the gap, with a mean of 4.46 but a success rate of 66.7%. Adding the two non-students lowers T4 from 4.23 to 4.13 (SD 1.30). The clash messages were rated fairly clear: among students 4.00 (SD 1.29) for class time (T4) and 4.15 (SD 1.07) for exam date (T5), and 3.93 and 3.87 among all 15, so the class-time message was the less clear one for students.

TABLE IV
SEQ RATINGS (1–5) BY TASK, MEAN (SD)

Task	Students (n = 13)	All respondents (n = 15)
T1	4.69 (0.63)	4.60 (0.74)
T2	4.38 (0.87)	4.33 (0.90)
T3	4.08 (1.19)	4.13 (1.13)
T4	4.23 (1.24)	4.13 (1.30)
T5	4.46 (1.20)	4.40 (1.18)
T6	4.69 (0.63)	4.60 (0.74)
T7	4.85 (0.38)	4.87 (0.35)

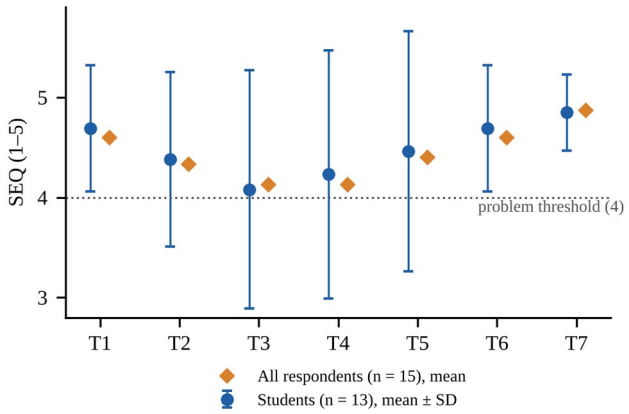


Fig. 3. SEQ rating per task. Error bars show ±1 SD and can extend beyond the limits of the scale.

The mean SUS score of the 13 students was 75.6 (SD 17.1; 95% CI 65.3–85.9), above the benchmark of 68, and 8 of the 13 (61.5%) scored above 68 (Fig. 4). The lower end of the interval is below 68, so with this sample size we can say only that the mean is above the benchmark, not that the true value is. For all 15 respondents the mean was 75.5 (SD 17.5; median 80.0; range 50–97.5); the two non-students averaged 75.0 (55.0 and 95.0). P02 answered 4 and P05 answered 5 to every item, which gives 50 in both cases and may not reflect a considered judgement; without these two the mean is 80.2 for the remaining 11 students and 79.4 for 13 respondents. Among the items (students, adjusted to 0–4), item 7, “most people would learn to use this system very quickly”, was highest (3.8) and items 1 and 3 followed (3.5). The lowest were item 10, needing to learn a lot before getting going (2.1), and item 4, needing the support of a technical person (2.2). Identical answers also push up the raw means of the negatively worded items, so differences between items should be read cautiously. Among favourite features (multiple choice) the separation of required and GenEd courses and the seat-colour labels were chosen most often (9 of 13 students each; 10 of 15 overall), followed by the review before confirmation (7; 8 overall), the history (7) and the class-time clash alert (6).

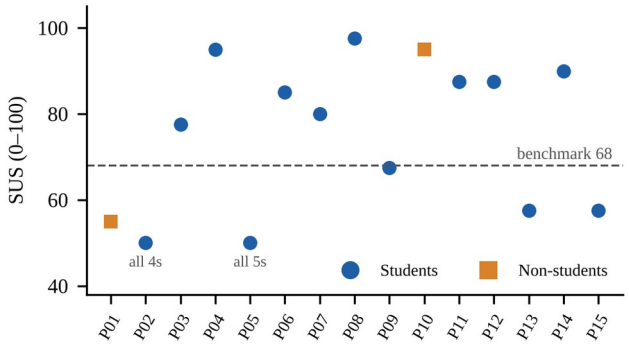


Fig. 4. SUS score per respondent (n = 15). The dashed line is the benchmark of 68.

C. Usability problems (RQ3)

Eight respondents (P01, P05, P06, P08, P09, P10, P12, P13) commented in the open-ended questions, and observation added further problems, giving eleven problems; Table V condenses them into eight rows by merging the two confirm-button problems. Two further comments are not in the table: P13 asked for a calendar view of the timetable (H2, H6) and P05 commented on the visual appearance (H8). Two non-students (P01, P10) contributed three of the table’s problems. Two tasks met the success criterion for trouble, T3 (60%) and T5 (66.7%); no task met the SEQ criterion and none the error criterion (the highest share of participants who erred was 40%, in T3). The elements behind T3 and T5 are problems 1 and 2, both about finding courses. Severity ratings were not assigned in this paper.

TABLE V
USABILITY PROBLEMS FROM OPEN-ENDED COMMENTS

#	Problem	Source	Screen element	Heuristic
1	GenEd course type (e.g., HyFlex) not shown; no skill-category filter; categories unclear.	P06, P08, P13; seen in T3	GenEd tab, filters	H6, H7
2	Academic English not found and not told apart from general English (T5).	Observed (5)	Course list	H2, H4
3	“Available seats” filter does not show what it does.	P09, P12	Seat filter	H1, H2
4	Class-time text too small; wants all courses in one menu.	P01, P13	Time text, course menu	H7, H8
5	Confirm button not found (T6) or too far down.	Observed (2); P10	Confirm button	H1
6	Exam date hard to see.	P10	Section card	H6, H8
7	History should summarise day and time per course and shade whole rows.	P06	History page	H1, H6
8	Required/GenEd switch should stand out more.	P08	Tab buttons	H1, H4

V. DISCUSSION

A. What worked

The SUS mean above 68, the high SEQ ratings and the clarity ratings of the clash messages suggest that participants found the prototype easy to use overall, although success in T3 and T5 was below 80%. The features they chose most often fit the design intent: the tabbed split of required and GenEd courses (H2, H4), and the review step and history (H3, H5), and T6 (4.69) and T7 (4.85) were rated easy. Estevez et al. [9] found that students confused “added to cart” with “registered”; our confirm step and success state address this, but T6 success was 86.7% and two participants could not find the confirm button, so we cannot yet claim that the confirm step removes the confusion. The clash alert was the least-chosen favourite (6 of 13 students). T4 was completed by all 15 participants with no errors, yet its message was rated 4.00 for clarity, so the alert worked but its wording can improve.

B. What to fix

Most problems concern the meaning and visibility of elements, not missing functions. Problems 1 and 2 lie behind the two tasks that missed the success criterion: users could not tell GenEd course types apart or find academic English among similar courses, which points to grouping, naming and labelling. Problem 3 is a labelling problem: the “available seats” chip does not tell users what it does or whether it is on, which is the kind of gap between what is seen and what is meant that icon studies describe [12], [14]. Problems 4, 6 and 8 concern class-time text, the exam date and the tab buttons being too small or too weak. Studies of

icons [11], [12] link small elements to slower and less accurate selection, so size and salience of these elements is a plausible cause, but we neither manipulated nor measured sizes and treat this as a hypothesis for the next iteration. Time pressure may matter too: complexity lowered accuracy in [13] only under the strictest time limit. Our sessions had no real deadline, whereas real registration is competitive, so problems may be underestimated here.

C. SUS items 4 and 10

Items 10 and 4 were lowest although item 7 was highest, which looks contradictory. Possible explanations are the two respondents with identical answers, a reading of both items as referring to registration rules in general rather than to the screen, and a lack of built-in help (H10). The data cannot separate these, so a follow-up question is needed in the next round.

D. Limitations

- **Sample.** Convenience sample of 13 students, 12 of whom used REG every semester, so newcomers are not represented. Two non-students outside the inclusion criterion are reported separately.
- **Success and errors.** Counts come from all 15 participants and cannot be split by group, and success was judged by the moderator. Per-task times were not recorded.
- **Time data.** Only totals, including thinking aloud, were timed. They are not yet matched with participant codes, so they cannot be linked to SUS or REG experience. Five values (4:24, 3:49, 4:37, 5:21, 12:25) were noted as min.sec and read as min:s.
- **Ratings and devices.** SEQ used five points and was collected after all tasks, so it reflects memory and overall impression and is near its ceiling. Two SUS respondents gave identical answers to all items. Participants used a computer, an iPad or a mobile phone; the number per device was not recorded, so results may differ by screen size.
- **No comparison.** REG itself was not tested, so we cannot say the prototype is better, and the suspected pain points remain hypotheses. Satisfaction in [10] was measured with a different instrument and is not comparable with our SUS score.
- **Prototype and problems.** Mock data and no real seat changes or crowding. Problems come from the open-ended comments of eight respondents and from observation, were not rated for severity, and problem 5 (confirm button) may be specific to the tested version.

VI. CONCLUSION

For RQ1, mean task success was 87.6% (T3 60%, T5 66.7%, T6 86.7%, the rest 100%), and students needed a mean of 4:37.1 min for all seven tasks. For RQ2, ratings were high: SEQ 4.08–4.85 and SUS 75.6, with the caveats above. For RQ3, T3 and T5 were the troublesome tasks by the success criterion (none met the SEQ or error criteria); the elements involved are the missing GenEd type labels and categories, the grouping and naming of academic English courses, the unclear seat filter and the hard-to-find confirm button.

Next, we will (1) add GenEd type badges, skill-category filters and clearer course categories; (2) separate academic English from general English courses; (3) relabel the seat filter (e.g., “show only sections with seats left”) and show whether it is active; (4) check that the confirm button is always visible; (5) enlarge the exam date and class-time text; (6) mark additions and withdrawals in the history with text or symbols as well as colour; (7) strengthen the tab buttons; (8) revise the wording of the class-time clash message; (9) consider a calendar view of the timetable; and (10) add short help text for filters and steps. A second round should time each task, record success and errors per participant group, ask SEQ right after each task on a seven-point scale, include users unfamiliar with REG, compare the prototype with REG on the same tasks in counterbalanced order, and re-test the revised design.

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